

number			
5(a)	An explanation that makes reference to the following points: - because haemophilia is {sex-linked / sex-linkage} (disorder) (1) - which means the {gene for haemophilia / gene for blood clotting / gene for factor VIII} is located on the X chromosome (1) - therefore males with the {faulty / recessive / haemophilia} allele will have haemophilia (1) - whereas females need to {be homozygous recessive / have two faulty alleles} to have haemophilia (1)	NB Max 3 marks if written in context of colour-blindness and not haemophilia ACCEPT X-linked ACCEPT if dominant and recessive alleles shown on X chromosomes in a diagram NB penalise recessive gene ONCE ACCEPT if clearly indicated on a diagram mutation ACCEPT if clearly indicated on a diagram	(4)

number			
5(b)(i)	185218 correctly divided by 37988 (1) 4.9 (1)	e.g. 5 / 4.87 / 4.874 / 4.8744 Bald answer of 4.9 = 2 marks Bald answer not to 2 sig figs correctly rounded = 1 mark	(2)

Question number	Answer	Mark
5(b)(ii)	The only correct answer is C <i>A is incorrect because the answer is 2.35×10^{-3}</i> <i>B is incorrect because the answer is 2.35×10^{-3}</i> <i>D is incorrect because the answer is 2.35×10^{-3}</i>	(1)

number			
5(b)(iii)	credit reasonable suggestion (1)	e.g. people are {dying / being born} all the time (young) people may not be diagnosed yet / won't know not everyone can be counted not everyone is tested / has access to tests not all cases recorded not everyone will know they have haemophilia IGNORE lie / ashamed / won't admit to it / asymptomatic	(1)

Question number	Answer	Additional guidance	Mark
5(c)	An explanation that makes reference to the following points: - as thrombin is an {enzyme / catalyst / catalyses} (1) - without thrombin fibrinogen will not be converted to fibrin (by thrombin) (1) - without fibrin there will be nothing to trap {platelets / blood cells} (1)	NB max 2 marks if no context ACCEPT less fibrinogen converted to fibrin ACCEPT fewer {platelets / blood cells} trapped	(3)